Quiz 4 revision

CS-321 Operating Systems. Chapters 7 and 8: deadlocks, main memory.

- Closed book, fifteen minutes, four short questions.
- Bring the slides from lecture 7 and 8 for the review session.
- Everything in this deck is examinable.

Chapter 7: Deadlocks

- Four necessary conditions: mutual exclusion, hold and wait, no preemption, circular wait.
- Resource allocation graph: a cycle means deadlock only when every resource has one instance.
- Banker's algorithm: safe state, safety sequence, request granting.
- Prevention versus avoidance versus detection and recovery.

Chapter 8: Main memory

- Logical versus physical addresses; the MMU and relocation register.
- Contiguous allocation: first fit, best fit, worst fit; external fragmentation.
- Paging: page table, TLB, effective access time.
- Segmentation and segmented paging.

Practice

- Run the banker's algorithm on the example from lecture 7 with a fifth process added.
- Compute the effective access time for a 90 percent TLB hit ratio and 100 nanosecond memory.
- Draw the resource allocation graph for the dining philosophers and find the cycle.